



Rajshahi University of Engineering & Technology

Rajshahi-6204, Bangladesh

Department of Electrical & Computer Engineering (ECE)

Course No.: ECE 4000

Title : CRISISCLIP-X: EXPLAINABLE CLIP-BASED MULTIMODAL DISASTER CLASSIFICATION

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Context

- ❑ Disaster-related social media posts often contain both images and text.[1][2]
- ❑ These posts may provide useful information about damage, affected people, rescue activities, and urgent needs.
- ❑ However, many posts are noisy, duplicated, unclear, or irrelevant.[3]
- ❑ Manual screening becomes slow and difficult during emergency situations.

Proposed Idea

- ❑ CRISISCLIP-X is an explainable CLIP-based multimodal framework for disaster classification.

Input

Image + Text

Tasks

Informativeness
Humanitarian
Severity

Purpose

Support faster
human review

Goal: to classify disaster-related social-media posts more accurately and support structured emergency information screening.

- ❑ During disasters, social media contains large volumes of image-text posts.
- ❑ Some posts are useful for emergency response, but many are irrelevant, noisy, duplicated, or unclear.
- ❑ Manual screening is slow and difficult during time-critical situations.[4]
- ❑ A multimodal AI system can help prioritize useful information for human review and situational awareness.



Need: fast, structured, explainable disaster information management

Paper 1: Analysis of Social Media Data using Multimodal Deep Learning for Disaster Response[13]

Method & Result

- ❑ Used CrisisMMD[8] image-text pairs for informativeness and humanitarian classification.
- ❑ Text branch: 5-layer CNN with word2vec; image branch: VGG16.
- ❑ Early feature fusion improved F1 to 84.2% for informativeness and 78.3% for humanitarian classification.

Limitations

- ❑ Static word2vec and shallow CNN have limited contextual understanding.
- ❑ Simple concatenation does not model strong cross-modal interaction.
- ❑ Limited XAI support and weak handling of minority humanitarian classes.

Relevance to this thesis: validated multimodal disaster learning and motivated stronger CLIP-based feature learning, imbalance handling, and XAI.

Literature Review (cont'd)

Paper 2 : Analysis of Multimodal Social Media Data Utilizing ViT Base 16 and GPT-2 for Disaster Response[9]

Method & Result

- ☐ Used ViT-B/16 for images and GPT-2 for text on CrisisMMD[8] informativeness classification.
- ☐ Combined unimodal predictions using Random Forest.
- ☐ Final multimodal model achieved 84.66% accuracy and ROC-AUC of 0.87.

Limitations

- ☐ Fusion was decision-level, so rich hidden image-text interaction was lost.
- ☐ Only binary informativeness was tested; humanitarian and severity tasks were not evaluated.
- ☐ No XAI and no comparison with CLIP-style joint vision-language models.

Relevance to this thesis: motivated feature-level multimodal fusion and direct CLIP-based image-text representation.

Paper 3 : Crisis-DIAS: Towards Multimodal Damage Analysis — Deployment, Challenges and Assessment[11]

Method & Result

- ❑ Proposed a gated three-stage pipeline for informativeness, infrastructure damage, and severity.
- ❑ Used RCNN/FastText for text and Inception-v3 for images with attention fusion.
- ❑ Reported high F1 scores and real-time processing of about 100 tweets/min.

Limitations

- ❑ Sequential gates can cause error propagation from earlier stages.
- ❑ Severity analysis mainly focused on infrastructural or utility damage.
- ❑ Limited XAI, missing-modality robustness, and cross-dataset validation.

Relevance to this thesis: inspired sequential disaster screening, but this thesis adds CLIP, XAI, MEDIC evaluation, and parameter-efficient adaptation.

Observed Gaps in Prior Studies

- ❑ Many earlier models relied on CNNs, shallow text encoders, or simple multimodal fusion.[5]
- ❑ Several studies focused on only one task, especially informativeness, rather than a full disaster-analysis pipeline.
- ❑ Limited attention was given to explainable AI for interpreting both image and text predictions.
- ❑ Cross-dataset validation and practical deployment were often weak or missing.
- ❑ Minority classes and class imbalance remained difficult, especially for humanitarian categories and damage severity.

How This Thesis Responds

- ❑ Uses CLIP ViT-B/32 as a stronger multimodal vision-language backbone.
- ❑ Covers three tasks together: informativeness, humanitarian category, and damage severity.
- ❑ Compares zero-shot prompting, adapter tuning, LoRA, and partial fine-tuning.[6]
- ❑ Integrates explainability through saliency maps, token attribution, and Gradient $\times$ Input.
- ❑ Evaluates generalisation on MEDIC and demonstrates deployment through a Streamlit application.

Based on the selected studies: Ofli et al. (2020), Urse et al., Gite et al. (2025), and Agarwal et al. (Crisis-DIAS, 2019).

Main gap: prior work did not combine a strong CLIP-based backbone, parameter-efficient adaptation, XAI, cross-dataset validation, and practical deployment in one unified framework.

Objectives

01

Develop a Multimodal Framework:

CLIP-based image-text disaster classification.

02

Perform Multi-task Analysis:

Informativeness, humanitarian category, and severity.

03

Compare Adaptation Methods:

Zero-shot, Adapter, LoRA, and partial fine-tuning.

04

Improve Reliability & XAI :

Class imbalance handling and explainable predictions.

05

Validate & Deploy

:CrisisMMD[8]/MEDIC evaluation and Streamlit prototype.

Research Goal: Efficient + Explainable + Deployable Multimodal AI for Disaster Information Classification

Two complementary datasets were used to develop and evaluate the proposed multimodal disaster-classification framework.

CrisisMMD[8]

Primary dataset for model development

Multimodal Twitter dataset containing disaster images and tweet text.

Collected from seven major natural disasters.

Supports informativeness, humanitarian-category, and damage-severity tasks.

Official splits support structured model training and evaluation.

MEDIC[15]

Secondary dataset for generalisation testing

Multimodal Dataset for Emergency and Disaster Image Classification.

Contains labels for informativeness, humanitarian category, damage severity, and disaster type.

Used for out-of-distribution and cross-dataset evaluation.

Helps test robustness beyond CrisisMMD.

Dataset role: CrisisMMD builds the main models; MEDIC checks cross-dataset robustness.

Events Covered

Crisis Event	Collection Period
Hurricane Irma	Sep 6–21, 2017
Hurricane Harvey	Aug 25–Sep 20, 2017
Hurricane Maria	Sep 20–Nov 13, 2017
California Wildfires	Oct 10–27, 2017
Mexico Earthquake	Sep 20–Oct 6, 2017
Iraq-Iran Earthquake	Nov 13–19, 2017
Sri Lanka Floods	May 31–Jul 3, 2017

Annotation Tasks

1

Task 1**Informativeness**

Informative / Not Informative

2

Task 2**Humanitarian**

Five aligned categories

3

Task 3**Damage Severity**

Severe / Mild / Little-or-no damage

CrisisMMD v2.0 multimodal baseline:**16,058 tweet texts and 18,082 images after cleaning**

CrisisMMD v2.0 includes task-specific distributions and requires label harmonisation for multimodal learning.

Informativeness	Image
Informative	9,374
Not Informative	8,708
Total	18,082

Humanitarian	Image
Affected Individuals	562
Infrastructure Damage	3,624
Not Humanitarian	8,708
Other Relevant Info	2,529
Rescue / Donation	2,231
Total	18,082

Damage Severity	Image
Little / No Damage	475
Mild Damage	839
Severe Damage	2,212
Total	3,526

alignment decision:

Semantically close minority categories were merged: injured/dead + missing/found → Affected Individuals; vehicle damage → Infrastructure and Utility Damage.

Selected file: task_informative_text_img_agreed_lab_dev.tsv

Split
Development (Dev)

Dataset Shape
1,573 rows $\times$ 9 columns

Task
Informativeness
(Text + Image)

Preview Summary

The file was loaded successfully using pandas with TSV format.

This split contains multimodal disaster samples with tweet text and image references.

The preview confirms 9 structured fields for analysis and model input.

First 5 rows were displayed during exploratory data inspection.

Columns

- event_name
- tweet_id
- image_id
- tweet_text
- image
- label
- label_text
- label_image
- label_text_image

This preview slide summarises the CrisisMMD v2.0 development split used for informativeness classification.

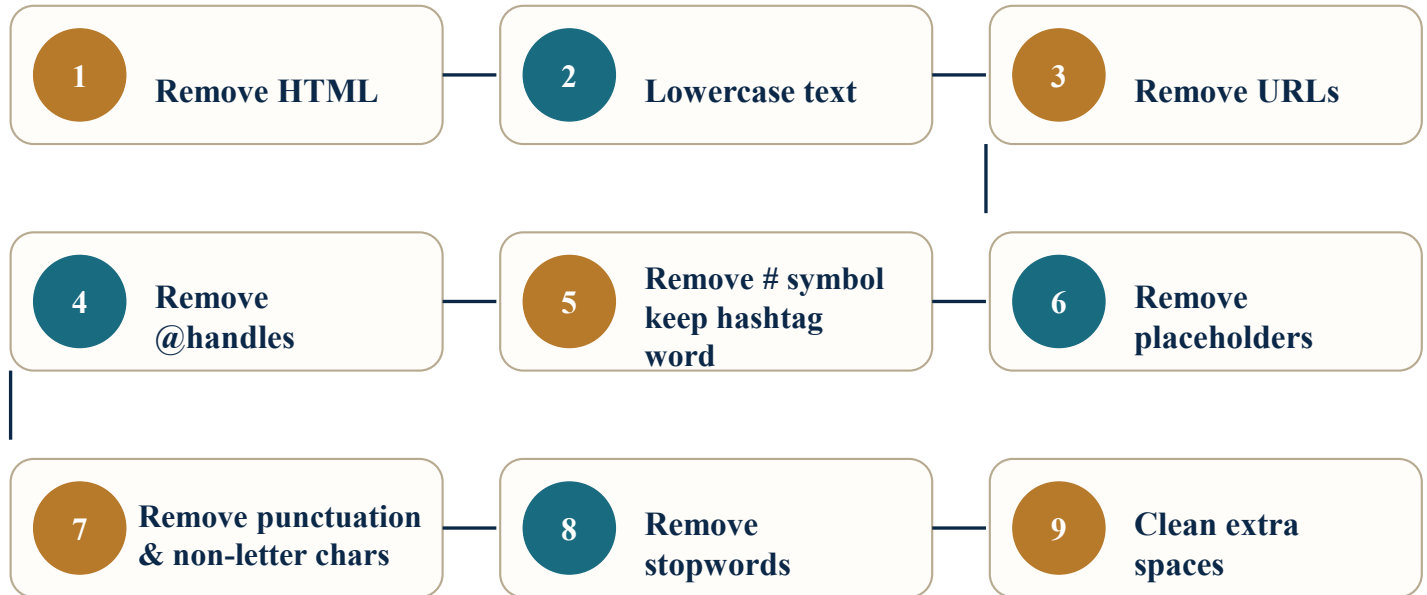
Text preprocessing steps followed in the implemented code

Input Preparation

Text was converted to string format.

HTML was removed using BeautifulSoup.

The text was converted to lowercase. URLs, Twitter handles, placeholders, punctuation, and non-letter characters were removed.[7]



Output: cleaned disaster-related tweet/caption text ready for CLIP tokenisation and model input.

Note: Hashtag words were preserved after removing the “#” symbol; stopwords were removed using NLTK English stopwords.

Proposed Methodology

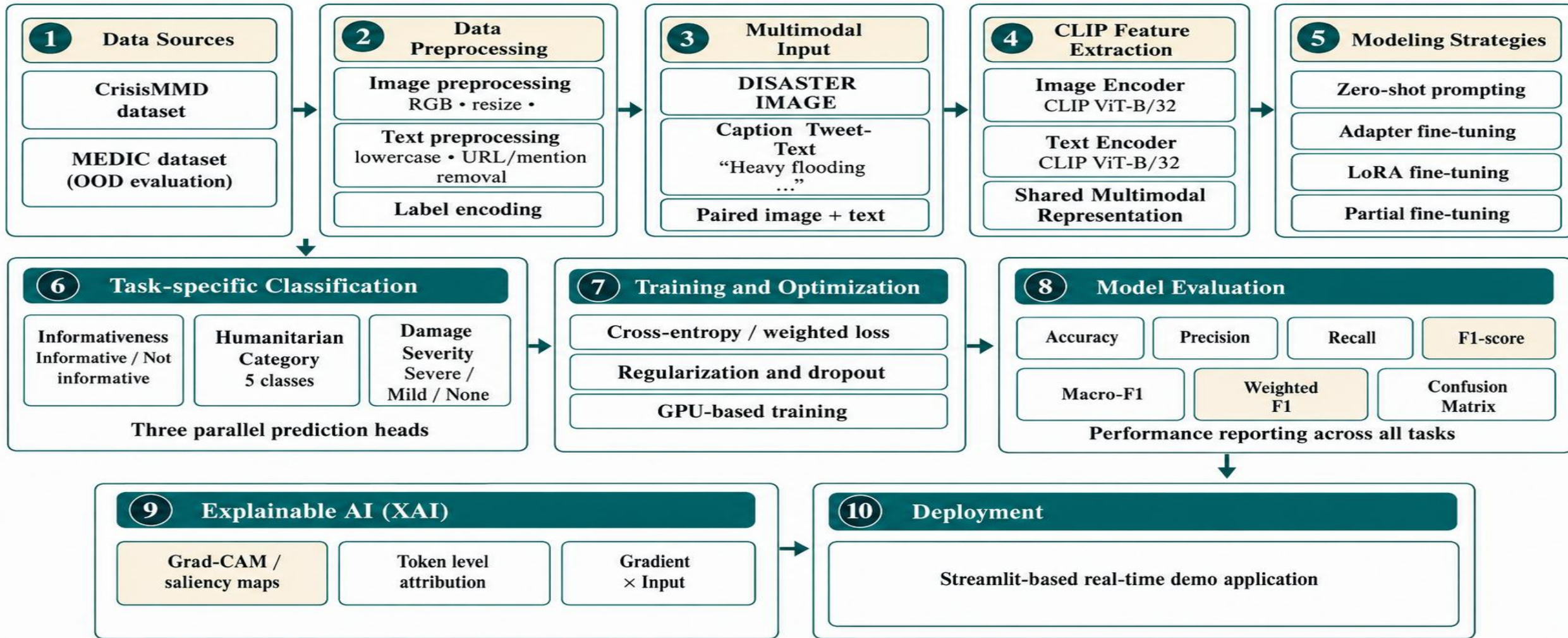


Figure : Proposed Methodology

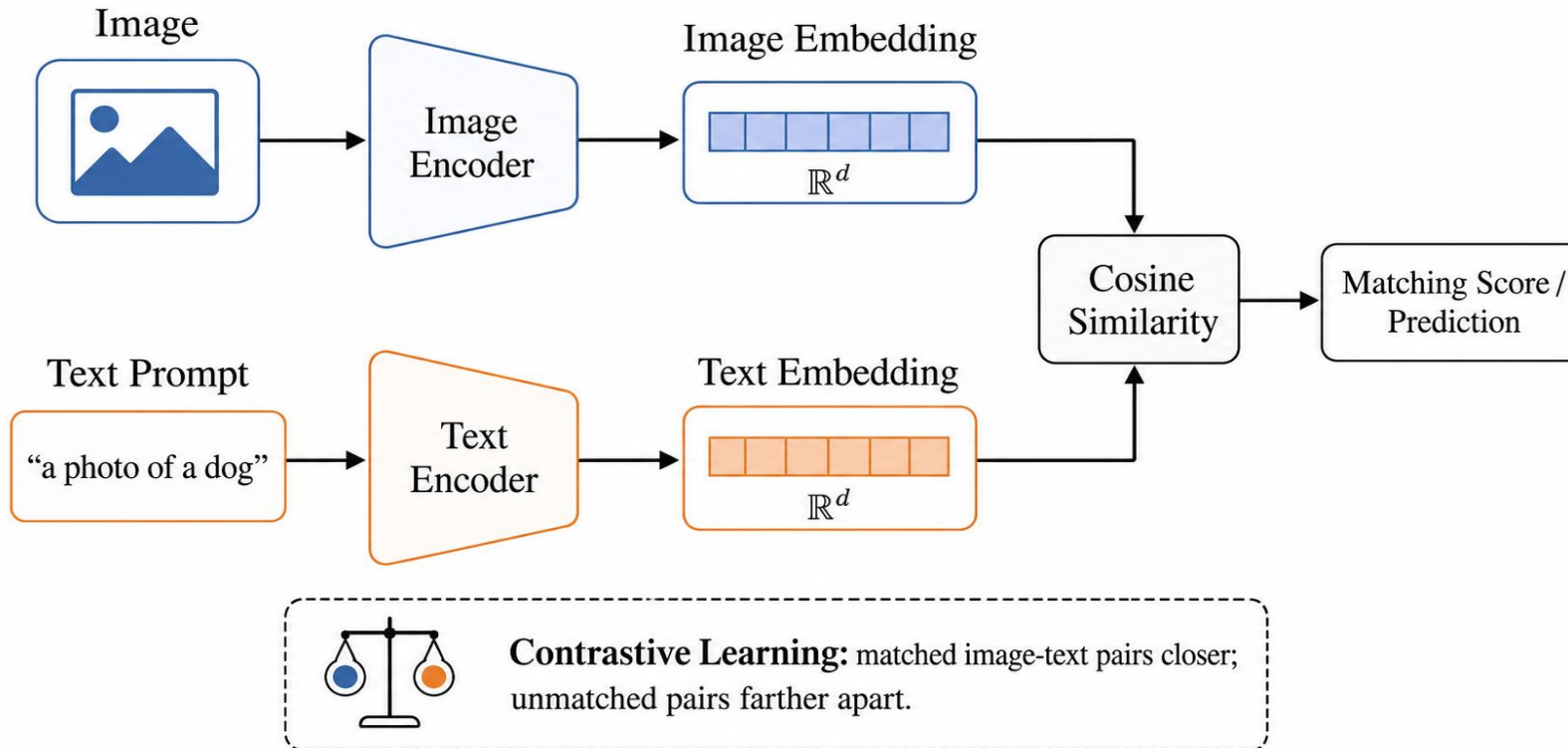


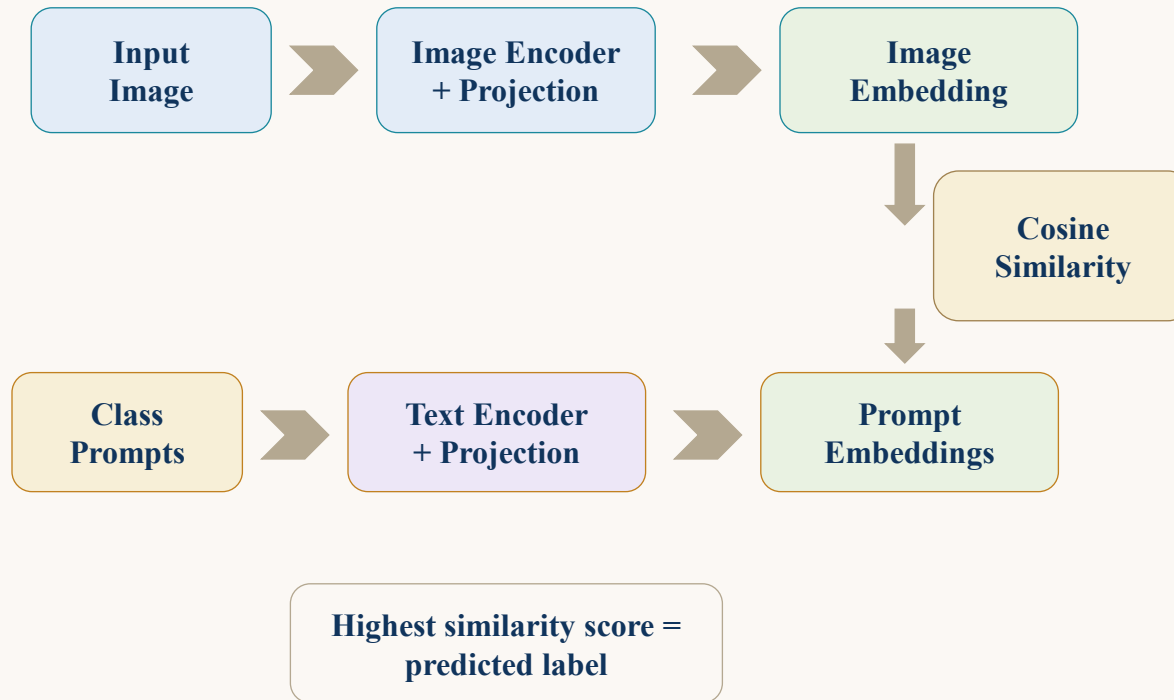
Figure : OpenAI CLIP Model Architecture

Why Choose CLIP?

Processes both images and text through separate encoders.
Projects both modalities into a shared semantic embedding space.
Supports zero-shot classification using text prompts.
Provides a strong pretrained vision-language backbone for disaster posts.
Well suited for multimodal social-media analysis where image and caption must be interpreted

Chosen as the shared backbone for all task variants.

Architecture Flow



Why Choose Zero-Shot?

Requires no task-specific training.

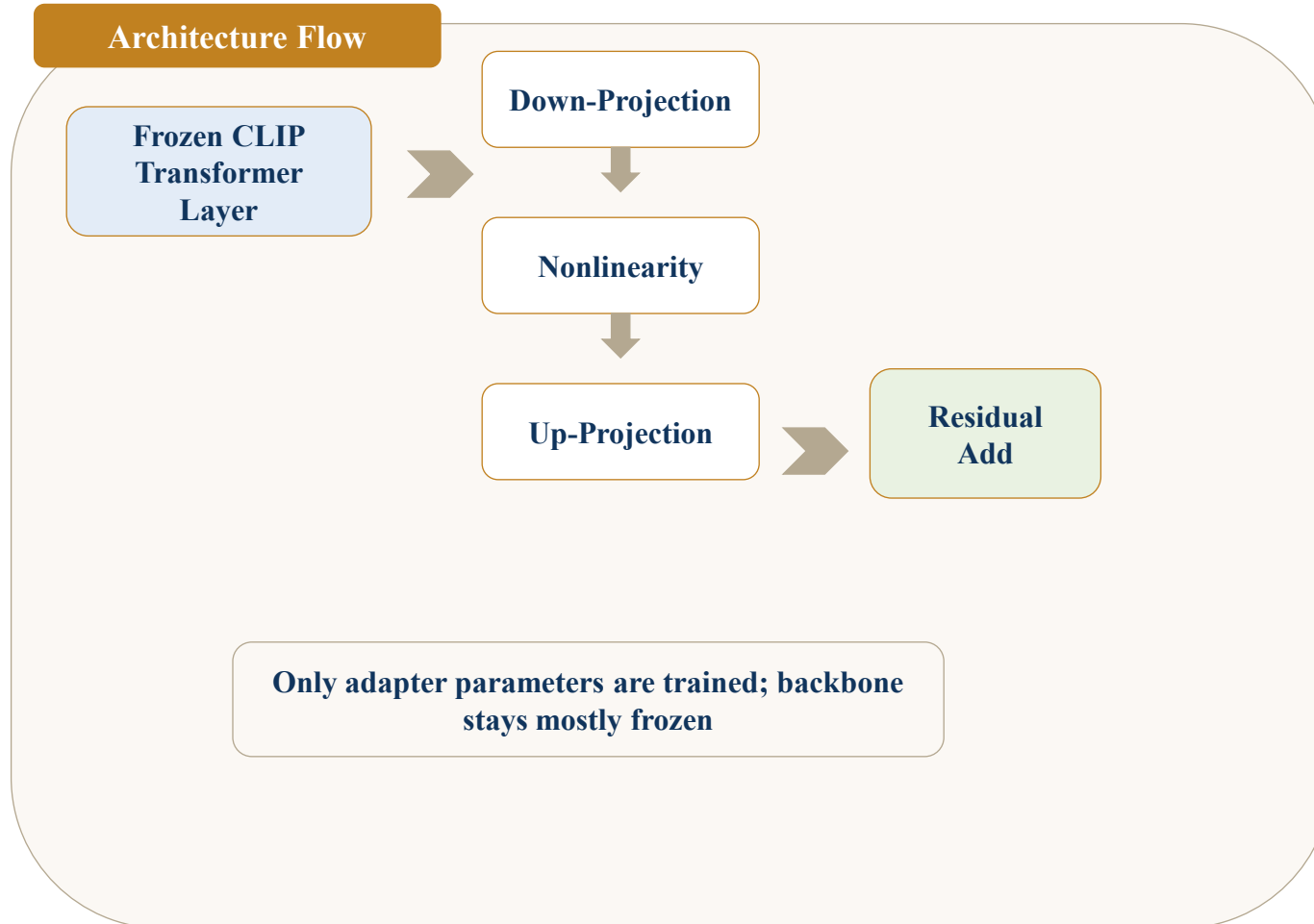
Uses only text prompts to define class labels.

Provides a clean baseline for comparison with fine-tuned models.

Computationally simple and easy to implement.

Helpful for checking whether pretrained CLIP knowledge already captures disaster concepts.

Adapter-based fine-tuning



Why Choose Adapters?

Insert small bottleneck modules inside transformer layers.

Train far fewer parameters than full fine-tuning.

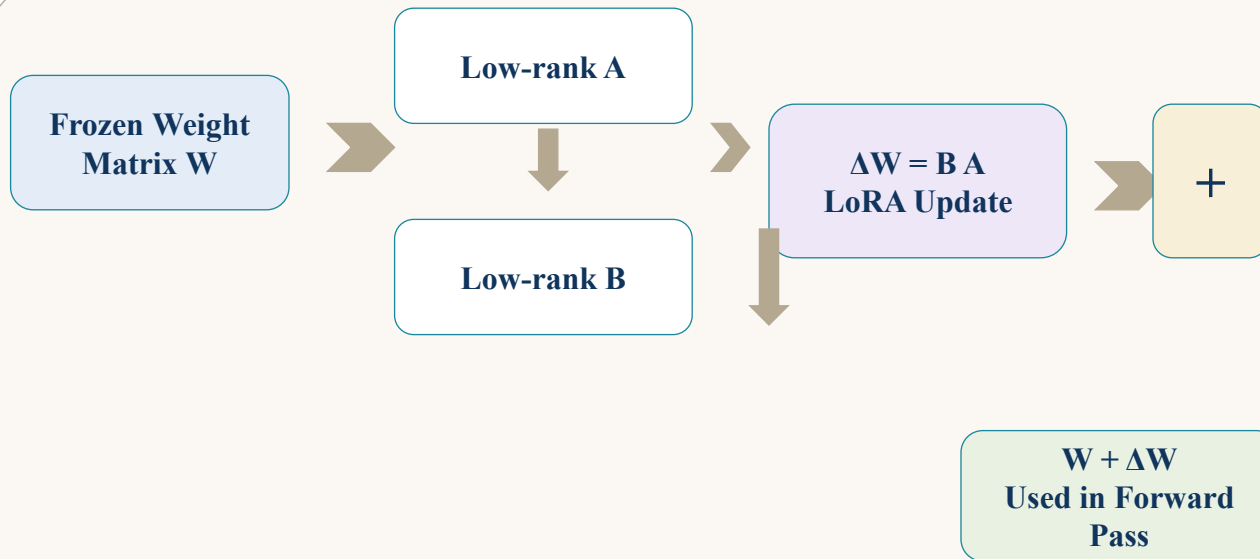
Preserve most pretrained CLIP knowledge while learning task-specific transformations.

Effective when training data are limited or classes are imbalanced.

In this thesis, adapters produced the best humanitarian-classification performance.

LoRA Implementation

Architecture Flow



LoRA is applied to attention projections such as q_proj, k_proj, v_proj, and out_proj

Why Choose LoRA?

Updates only small low-rank matrices instead of the full weight matrix.

Very parameter-efficient and storage-efficient.

Maintains strong performance while reducing computational cost.

Useful for deployment-oriented systems and repeated experimentation.

In this thesis, LoRA gave a strong balance between performance and efficiency, especially for severity classification.

Research Contributions

- CLIP ViT-B/32 framework integrating disaster images and text
- Adapter and LoRA fine-tuning with fewer trainable parameters
- Multi-task analysis: humanitarian category, informativeness, and severity
- Explainability through Grad-CAM and token-level attribution

Practical Applications

- Automatic analysis of disaster-related social media content
- Support for NGOs and rescue agencies in identifying urgent needs
- Rapid damage-severity assessment for faster decisions
- Interpretable predictions for AI-assisted decision systems

Societal Benefits

- Faster disaster management through reduced manual screening
- Improved prioritization of rescue and relief resources
- Quicker identification of affected and vulnerable populations
- Greater confidence through transparent, responsible AI

Usefulness & Future Potential

- Balances performance, efficiency, and explainability
- Real-time disaster monitoring platforms
- Humanitarian decision-support and edge-based analysis
- Future integration with emergency response networks

“CRISISCLIP-X enables efficient, transparent, and practical multimodal AI-based disaster understanding for supporting faster and more reliable humanitarian response.”

Model Configuration

BACKBONE MODEL

CLIP ViT-B/32

- Pre-trained vision–language model
- Image encoder + text encoder
- Image input: 224×224
- Maximum text length: 77 tokens

TRAINING FRAMEWORK

- PyTorch
- HuggingFace Transformers
- CUDA GPU acceleration

Training Parameters

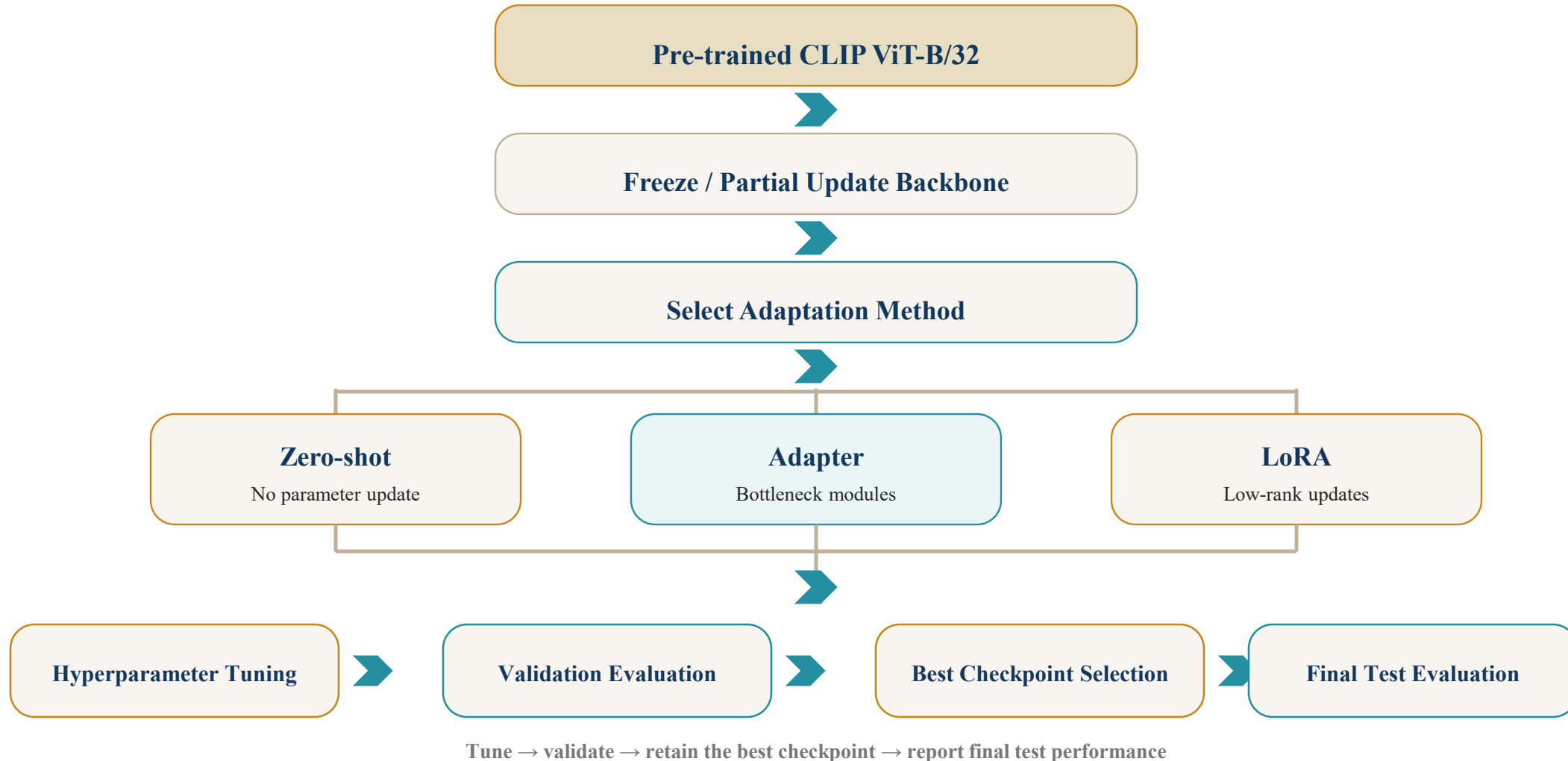
Optimizer	AdamW
Weight Decay	0.01–0.05
Loss Function	Cross Entropy
Label Smoothing	0.1
Scheduler	Warmup + LR Scheduler
Gradient Clipping	1.0

Fine-tuning Strategy

- 1 Zero-shot Prompting**
No trainable parameters
CLIP knowledge reused directly
- 2 Adapter Fine-tuning**
Small bottleneck modules
Lower memory; efficient adaptation
- 3 LoRA Fine-tuning**
Low-rank weight updates
Fewer trainable parameters

Preserve CLIP pretrained knowledge while improving disaster classification with minimum computational cost.

RESEARCH WORKFLOW



Dataset Configuration

DATASETS

CrisisMMD + MEDIC

Modalities: Image + Text

CLASSIFICATION TASKS

- 1 Informativeness Classification
- 2 Humanitarian Classification
- 3 Damage Severity Assessment

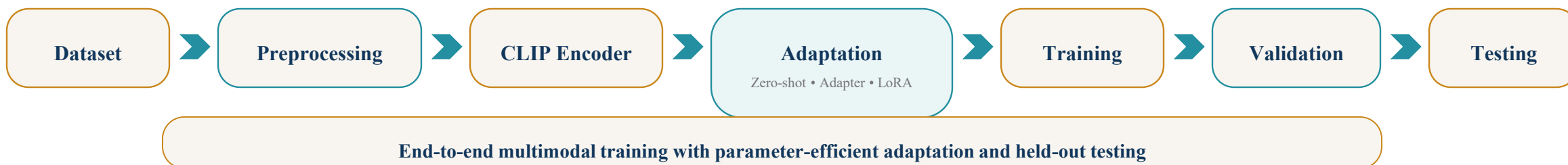
Computational Environment

Operating System	Microsoft Windows
Python Environment	Anaconda (tweet project)
CPU	Intel Core i5-12500H (12th Gen)
GPU	NVIDIA GeForce RTX 3050 Laptop
GPU Memory	4 GB
System Memory	16 GB
Storage	Samsung NVMe SSD 512 GB

Implementation Framework

- 1 Deep Learning
PyTorch
- 2 Multimodal
Hugging Face Transformers
- 3 Efficient Fine-tuning
PEFT
- 4 Deployment
Streamlit

COMPLETE WORKFLOW



This section presents quantitative performance, visual analysis, explainability, and comparative findings across the three classification tasks.

Tasks Evaluated

Informativeness classification
Humanitarian-category classification
Damage-severity classification

Evidence Used

Confusion matrices and ROC curves
t-SNE visualisation and example predictions
Grad-CAM, attention rollout, token attribution, and SHAP-style analysis

Result Trend

Fine-tuned models consistently outperformed zero-shot baselines.
Adapter tuning was strongest for humanitarian analysis.
LoRA remained a strong efficiency-performance trade-off.

Main quantitative findings

Informativeness
89.36%

Best accuracy with partial fine-tuning

Humanitarian
90.58%

Best accuracy with adapter fine-tuning

Severity
71.27%

Best accuracy with adapter fine-tuning

Informativeness

Core Performance

Best Accuracy

89.36%

Partial fine-tuning

LoRA Accuracy

87.17%

Efficient variant

ROC-AUC

0.9593

Strong separation

Normalized Confusion Matrix

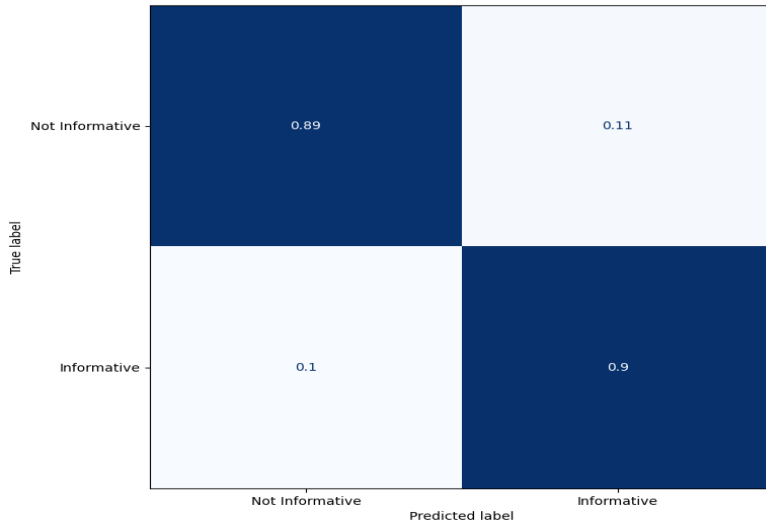


Figure : Normalized Confusion Matrix of Informativeness

Receiver Operating Characteristic (ROC) Curve

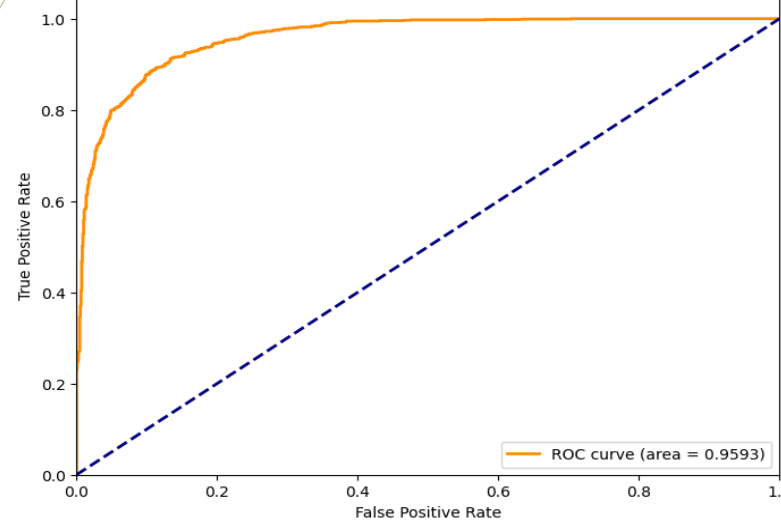


Figure : ROC Curve of Informativeness

What this shows

Both classes show strong recall: Not Informative ≈ 0.89 and Informative ≈ 0.90 .

The high ROC-AUC indicates that the model separates informative and non-informative posts reliably.

This task works well as the first gate before humanitarian and severity analysis.

Informativeness

Embedding Space and Sample Predictions

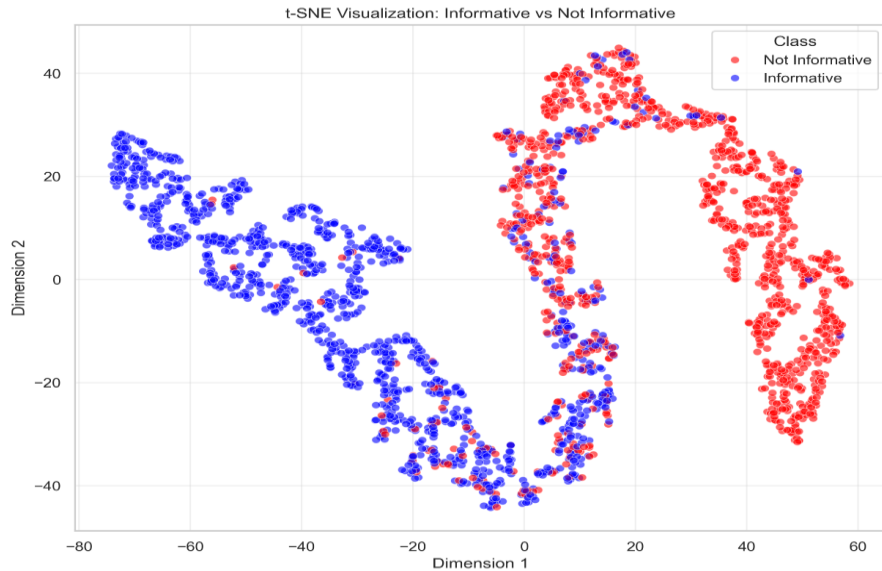


Figure:t-SNE Visualization of Informativeness

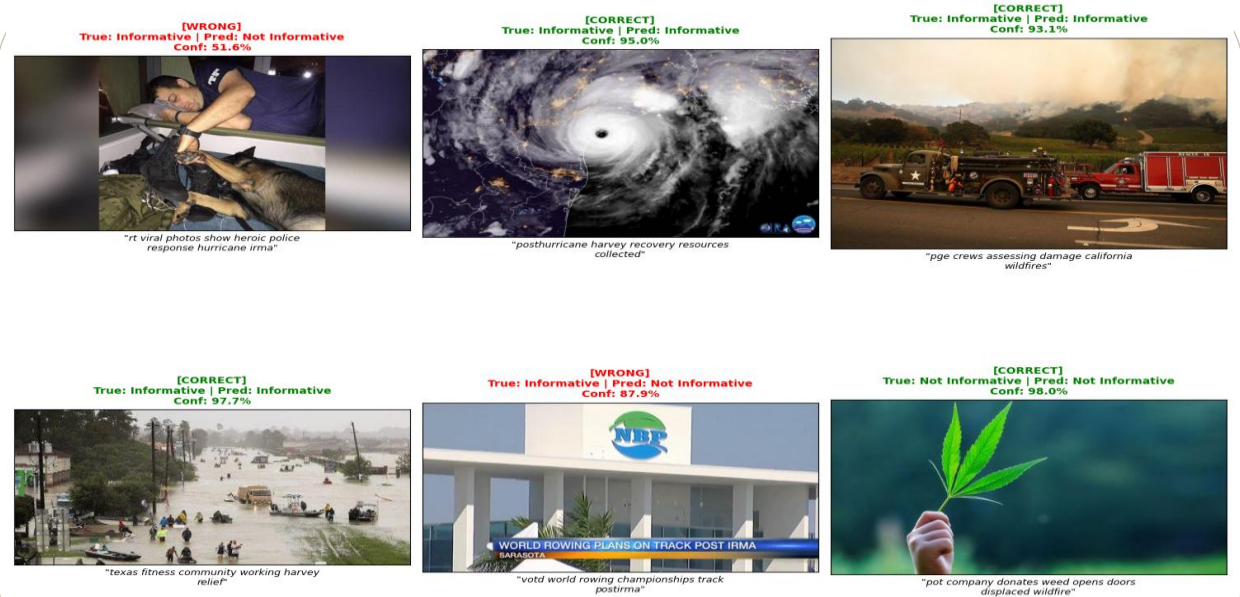


Figure : Representative Predictions of Informativeness

Interpretation

The embedding space shows clear separation between Informative and Not Informative examples, but overlap remains for ambiguous posts. Wrong examples usually involve weak image-text alignment, misleading captions, or visually non-disaster content.

Informativeness

Explainability Evidence

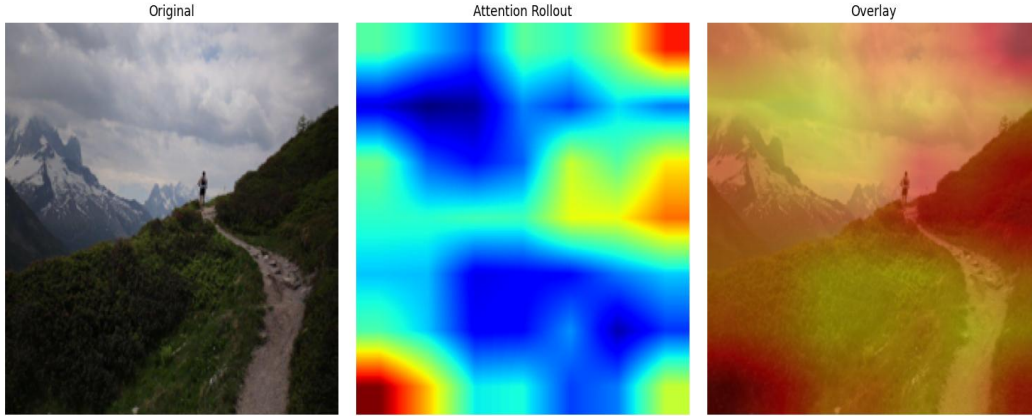


Figure: Grad-Cam Attention Rollout

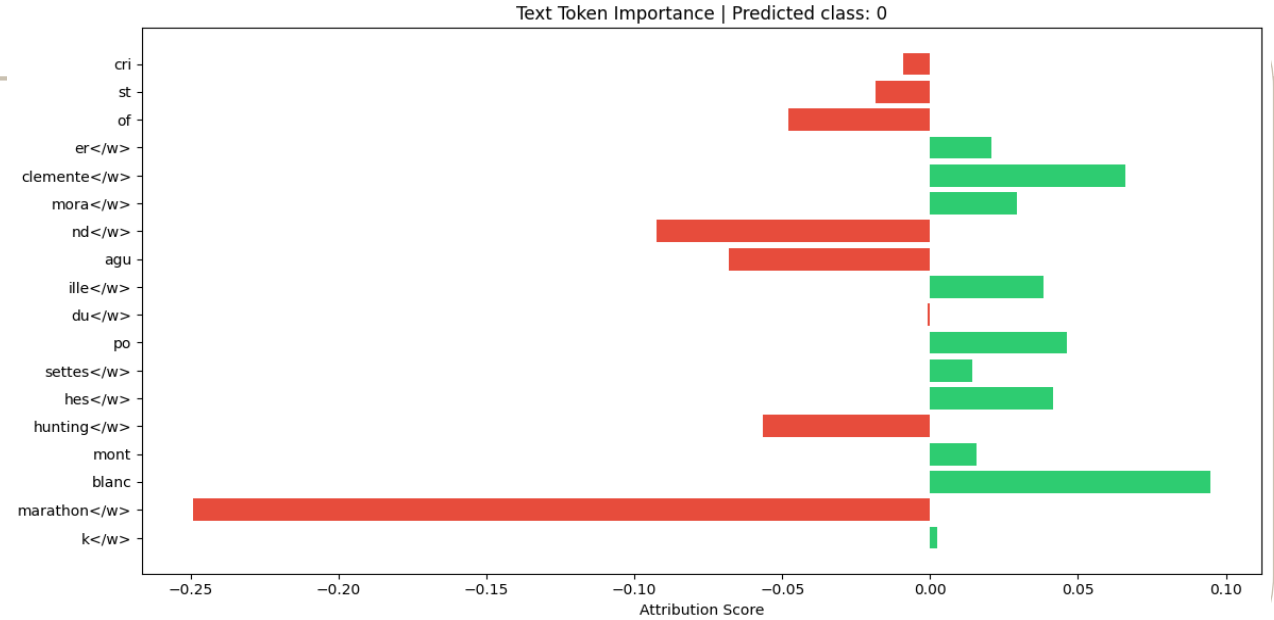


Figure :Text Token Importance of Informativeness

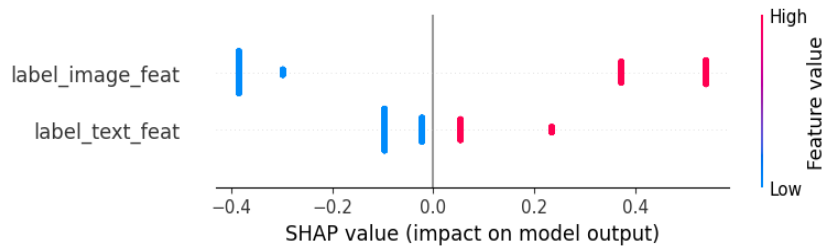


Figure : Modality-Level SHAP

Main message

XAI confirms that both modalities contributed to the informativeness decision.

Visual attention highlights image regions that influenced the decision, while token importance shows the text cues that pushed predictions toward or away from the target class.

These explanations support human review but should not be treated as proof of correctness.

Results & Analysis(cont'd)

Economic Feasibility & Cost-effectiveness For Informativeness

Performance Comparison

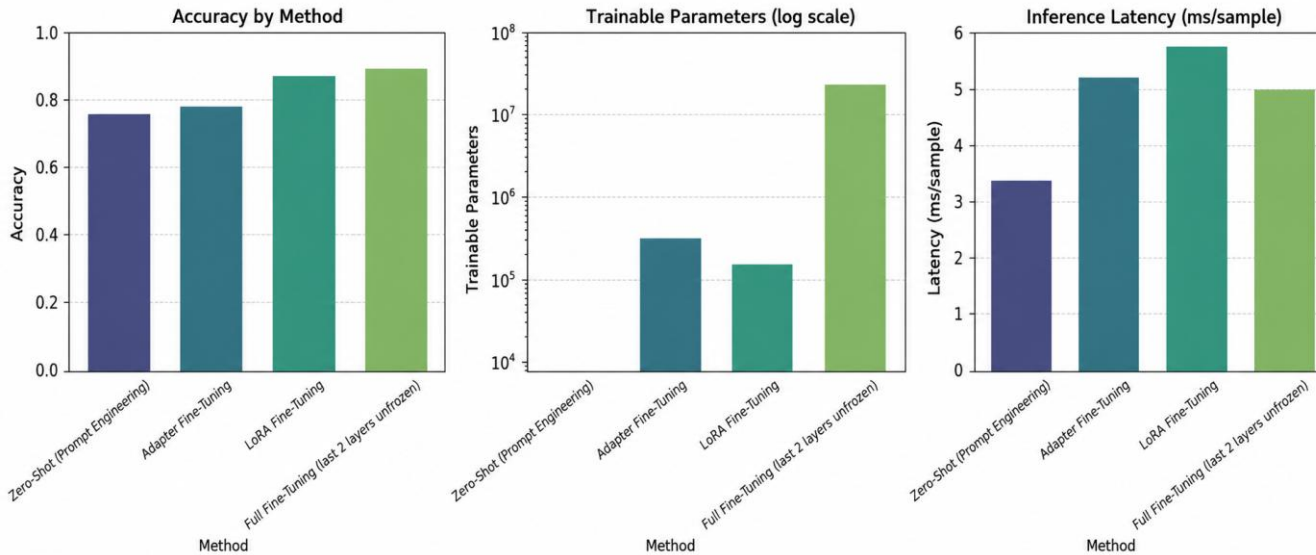


Figure : Performance Compare



Full fine-tuning achieved the highest informativeness accuracy.
LoRA provided strong accuracy with lower trainable-parameter cost than full fine-tuning.
Zero-shot inference was fastest and had the highest throughput.
VRAM usage stayed almost constant across zero-shot, adapter, and LoRA.

Computational Resource Comparison

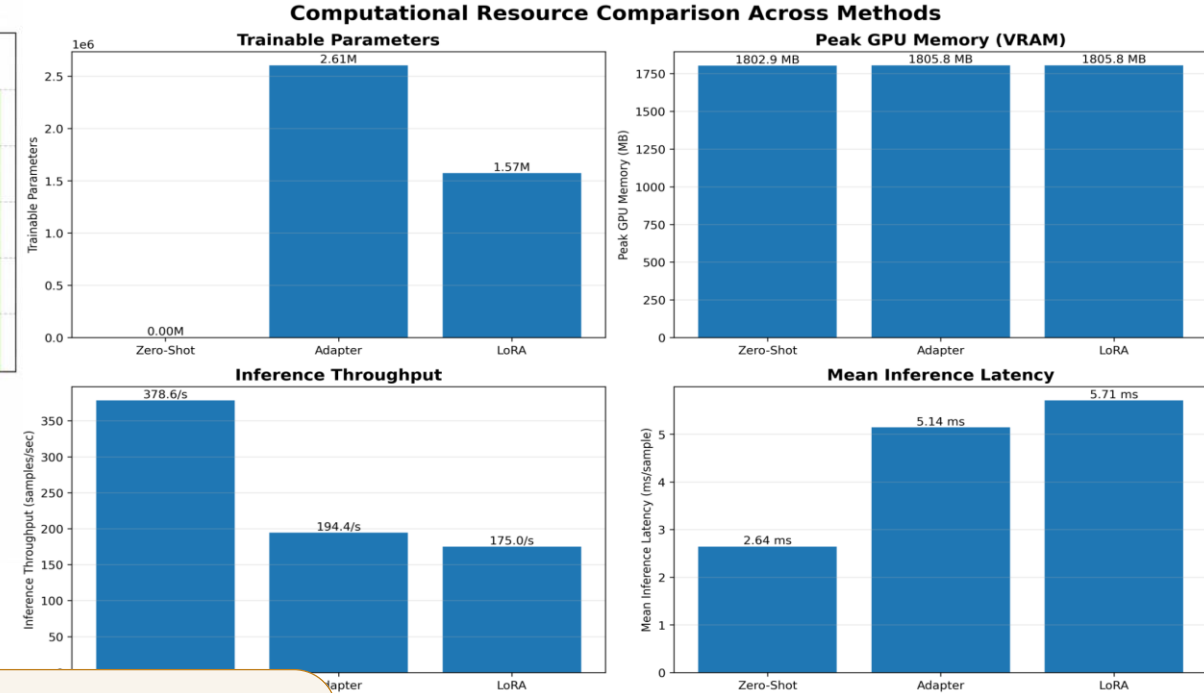


Figure : Inference count

Result

Fine-tuning improved performance, while LoRA offered the best performance-efficiency trade-off.

Humanitarian

Method Comparison and Best Model

Adapter Accuracy

90.58%

Best model

Adapter Macro-F1

83.24%

Best balance

Adapter Params

5.30M

3.39% trainable

One-vs-Rest ROC Analysis (Humanitarian Task)

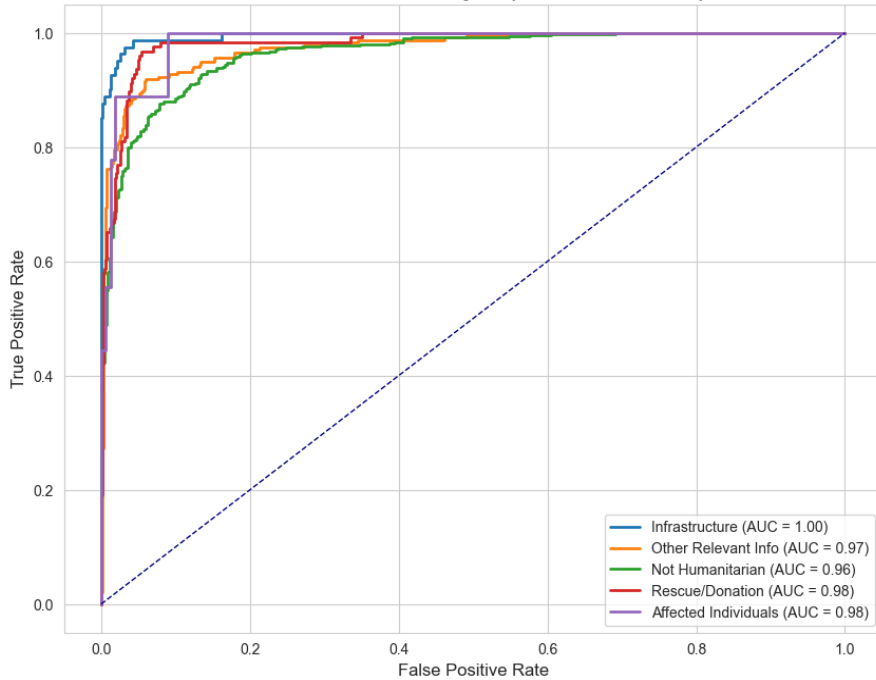


Figure : ROC curve of Humanitarian Classify

Normalized Confusion Matrix — ADAPTER

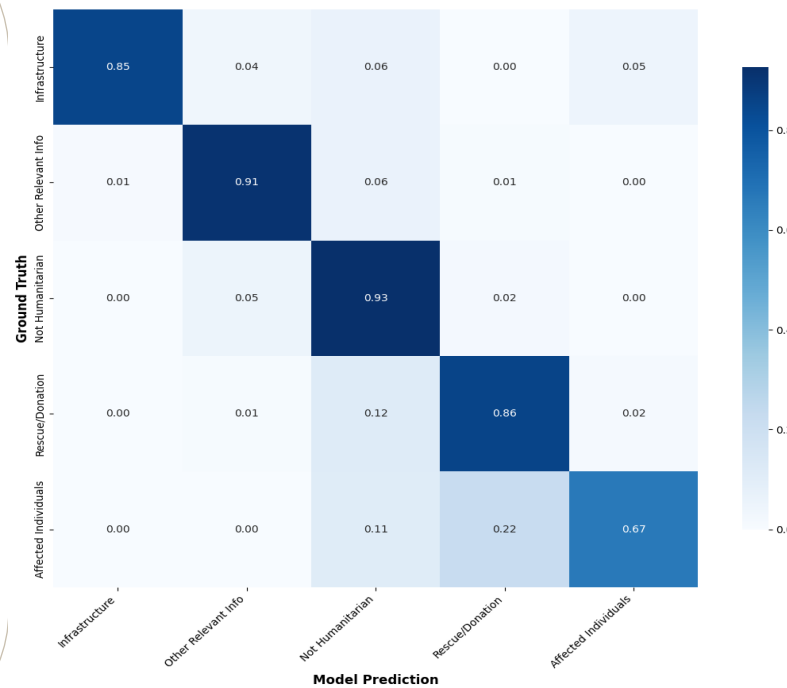


Figure : Adapter Confusion Matrix Humanitarian Classify

Normalized Confusion Matrix — LORA

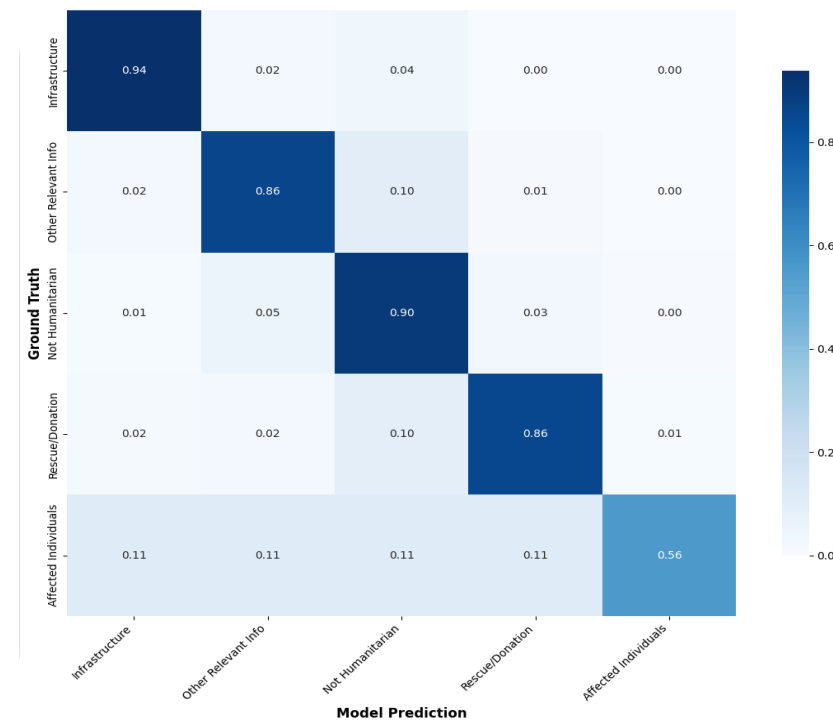


Figure : LoRA Confusion Matrix Humanitarian Classify

Humanitarian

Predictions and Class Separability

Qualitative Analysis: Per-Class Multimodal Predictions

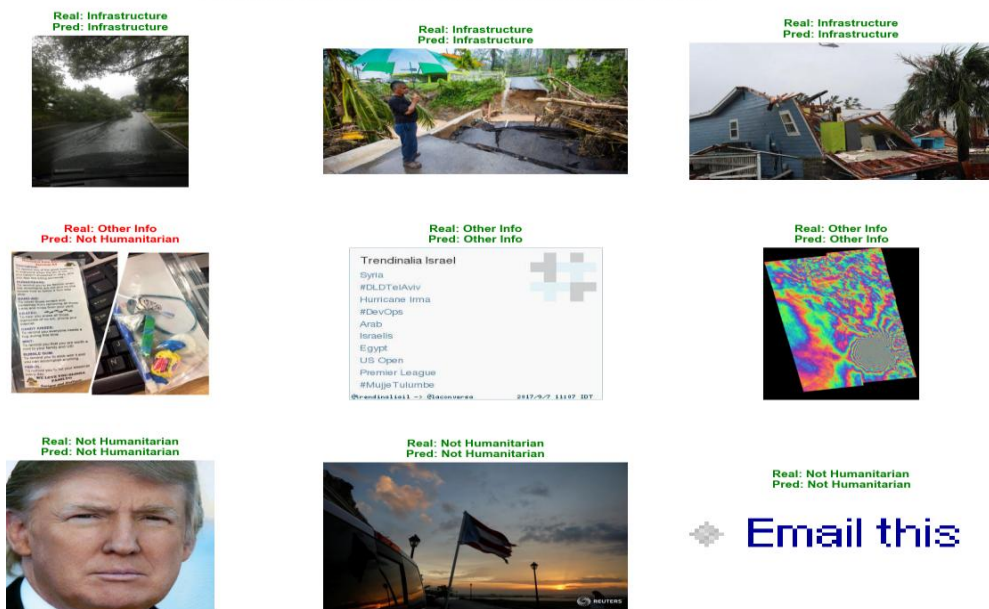


Figure : Representative Predictions Humanitarian Classify

t-SNE Projection of Fused Multimodal Embeddings

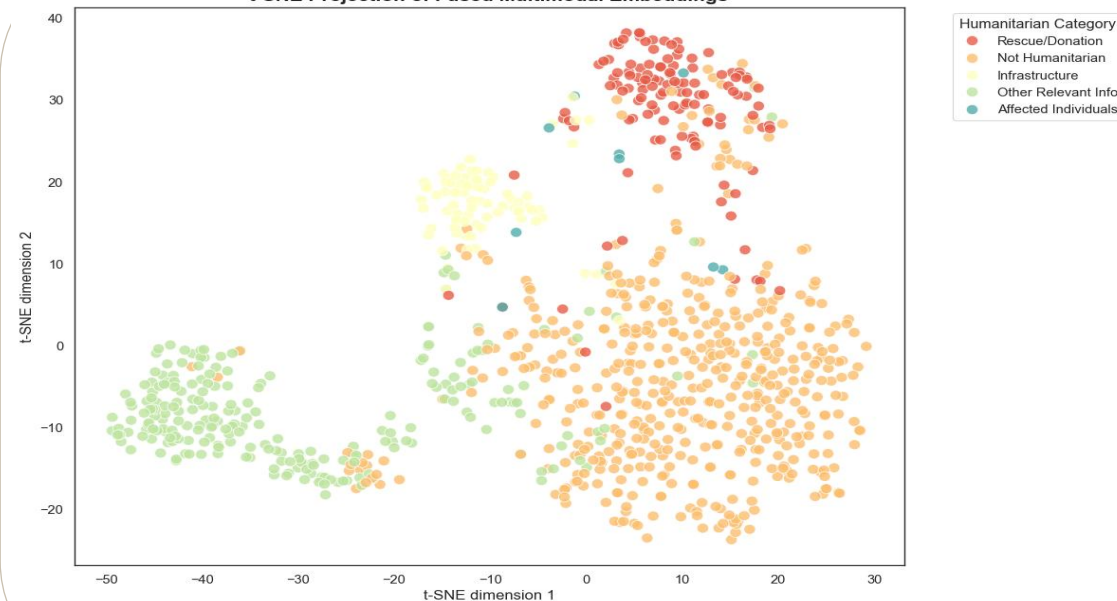


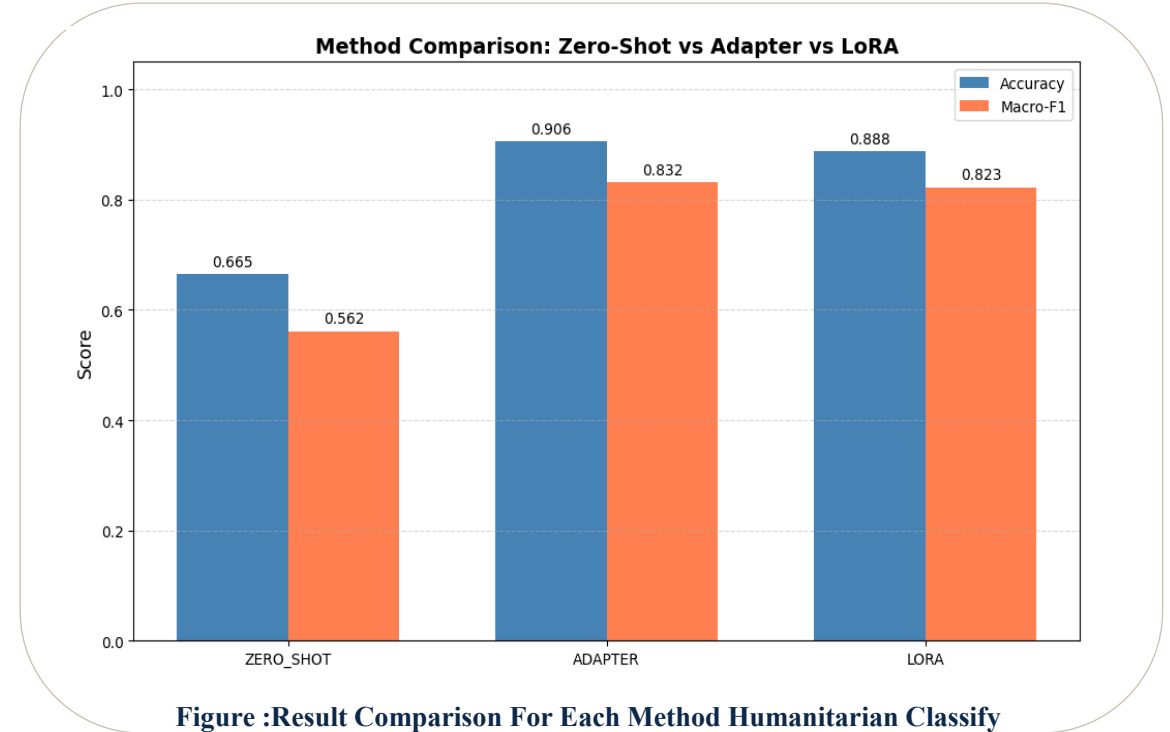
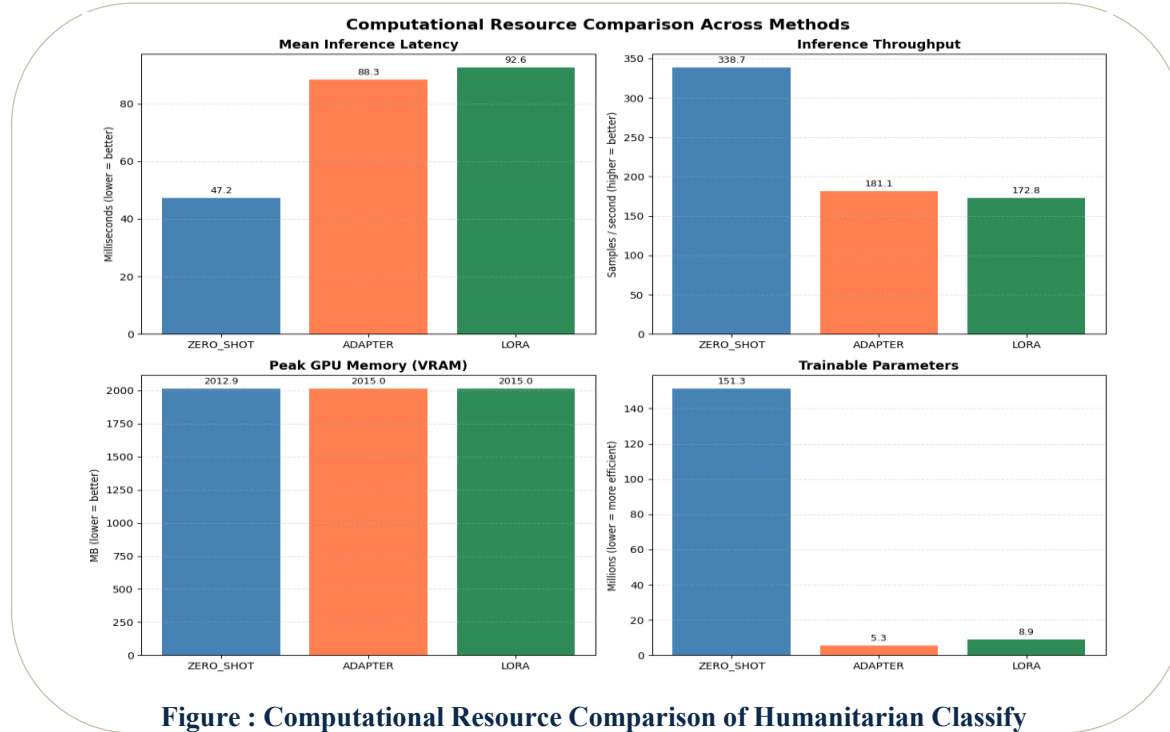
Figure :One-vs-Rest ROC Analysis Humanitarian Classify

Why this matters

Adapter improves performance, while LoRA remains attractive for deployment because it updates fewer parameters than larger fine-tuning approaches. ROC curves show strong class separability for major humanitarian categories, supporting the reliability of the classification pipeline.

Humanitarian

Economic feasibility/Cost-effectiveness



Why this matters

Adapter improves performance, while LoRA remains attractive for deployment because it updates fewer parameters than larger fine-tuning approaches. ROC curves show strong class separability for major humanitarian categories, supporting the reliability of the classification pipeline.

Humanitarian

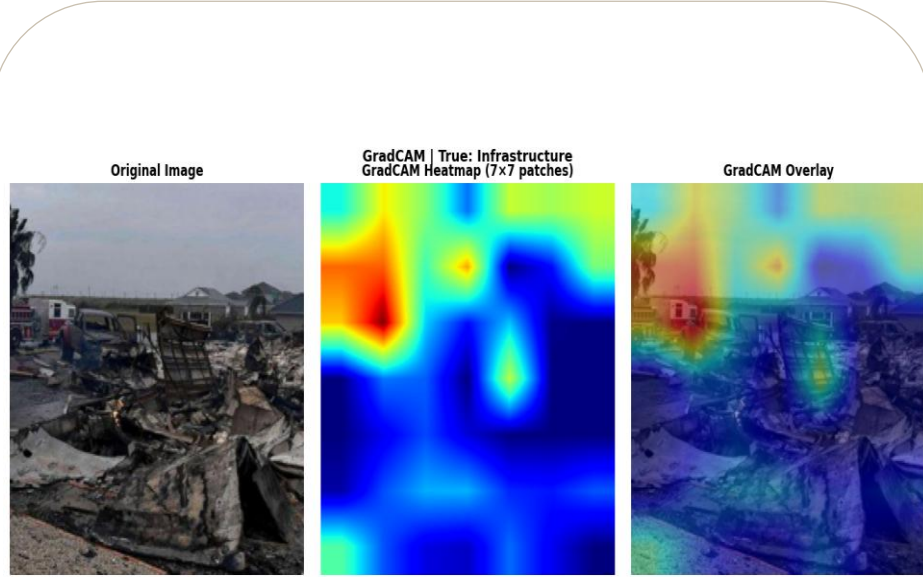


Figure : Grad-CAM Explanation Humanitarian Classify

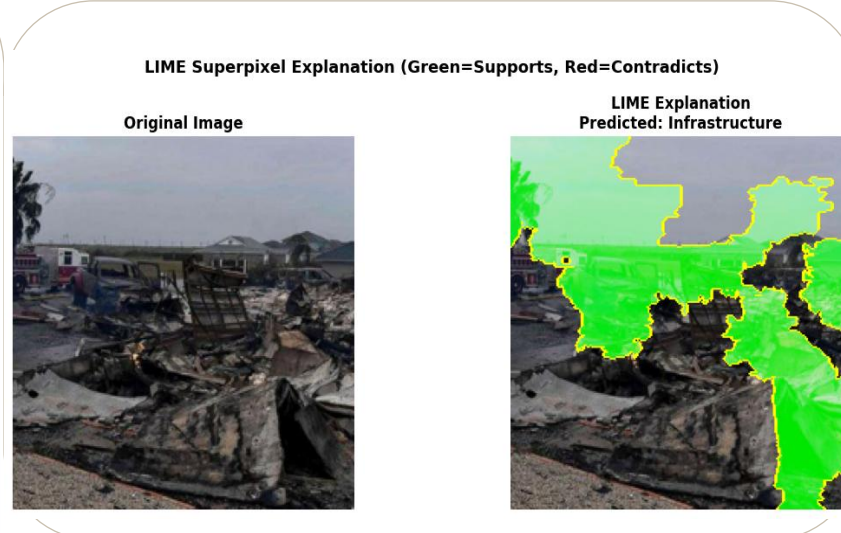


Figure : LIME Superpixel Explanation

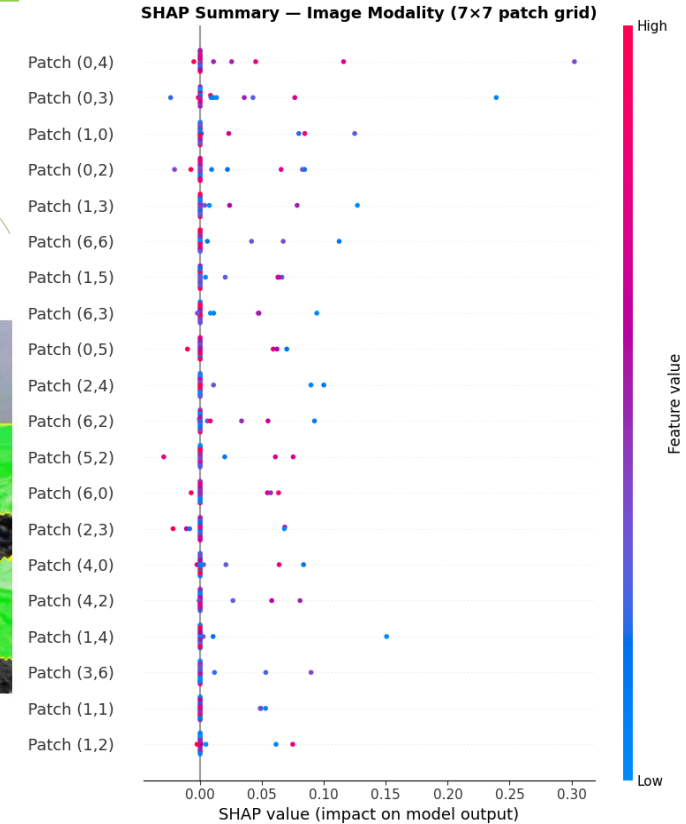


Figure : Image-Modality SHAP

Grad-CAM and LIME show that the model focused on damaged infrastructure and visually meaningful regions for humanitarian prediction.

SHAP-style image analysis indicates that certain image patches contributed more strongly to the output.

Together, these explanations make humanitarian classification more transparent for human reviewers.

Severity

Core Performance

Best Accuracy

71.27%

Adapter

LoRA Macro-F1

58%

Balanced classes

Weighted F1

69%

LoRA

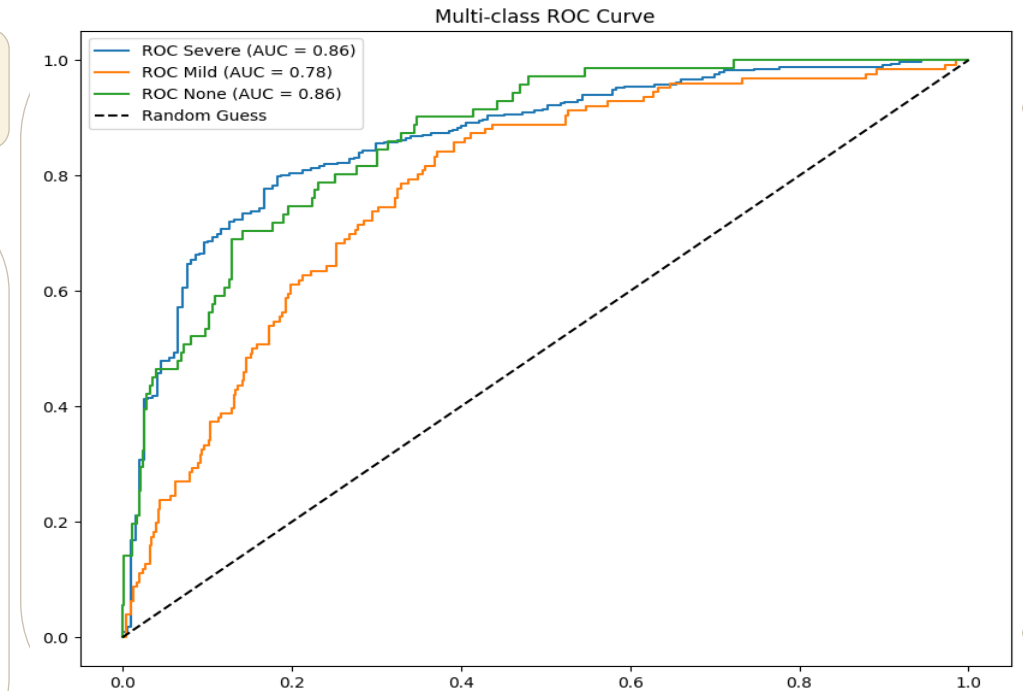
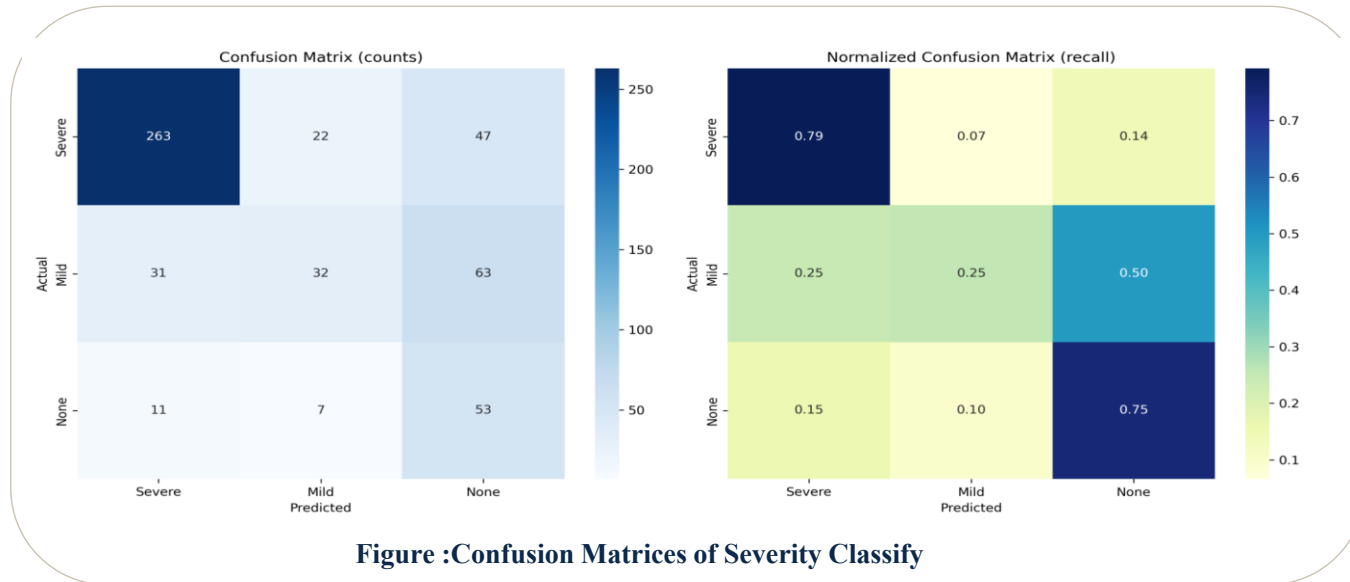


Figure : Multi-class ROC Curve of Severity Classify

Severity classification is harder because Mild damage overlaps visually with Severe and None.

ROC results show stronger discrimination for Severe and None than for Mild, which remained the most difficult severity class.

Severity

Method Trade-off and Qualitative Predictions

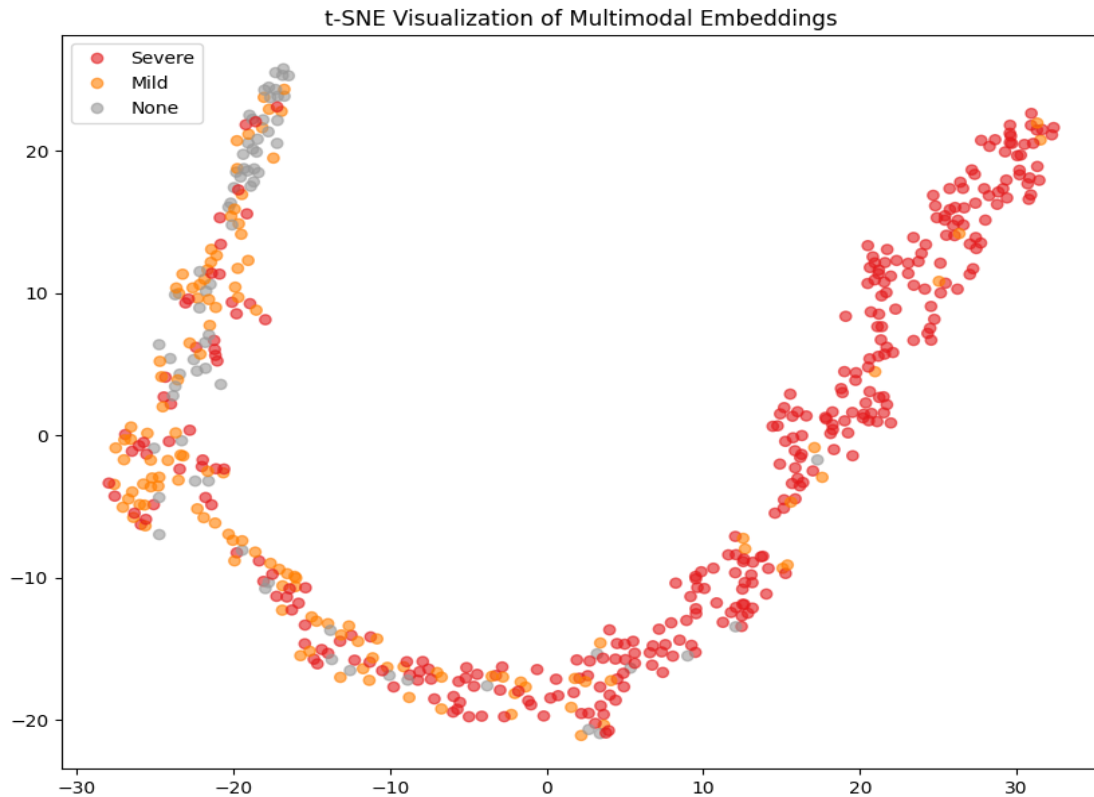


Figure:t-SNE Embedding of Severity Classify

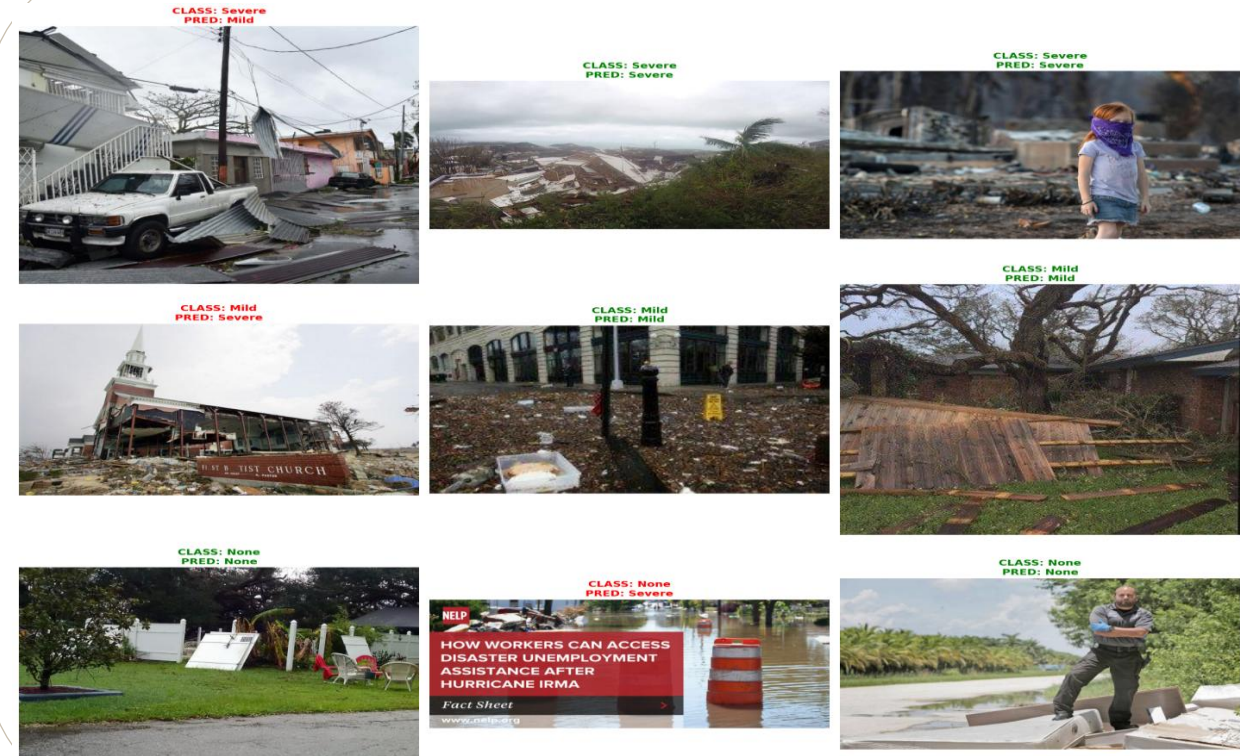


Figure :Example Predictions of Severity Classify

Severity

Economic Feasibility/Cost-effectiveness

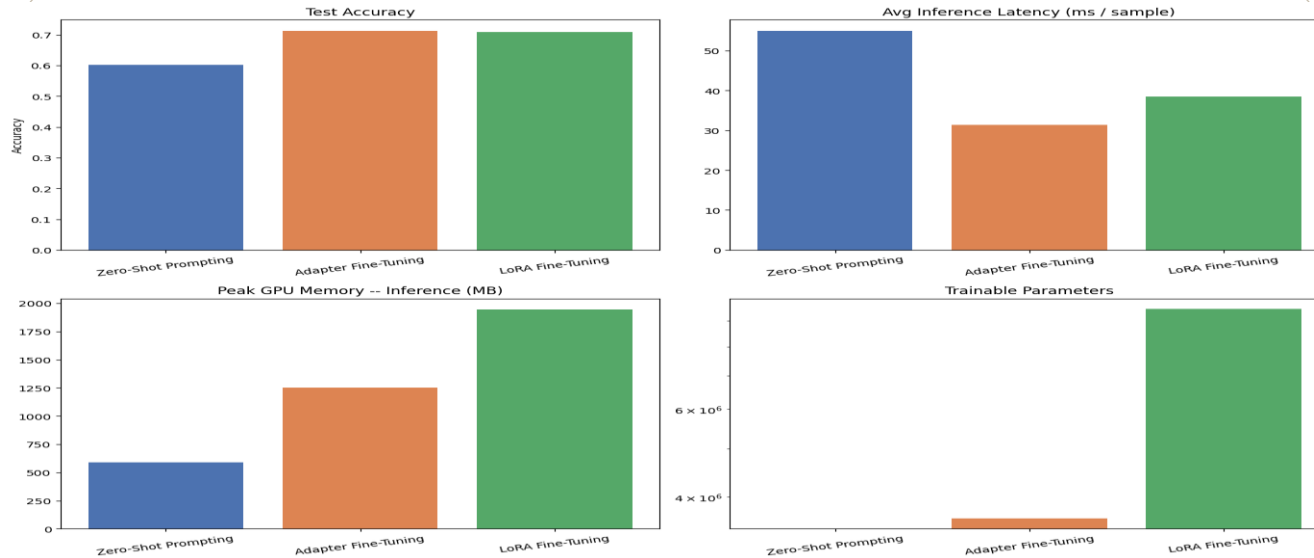


Figure :Accuracy, Latency, Memory, Parameters

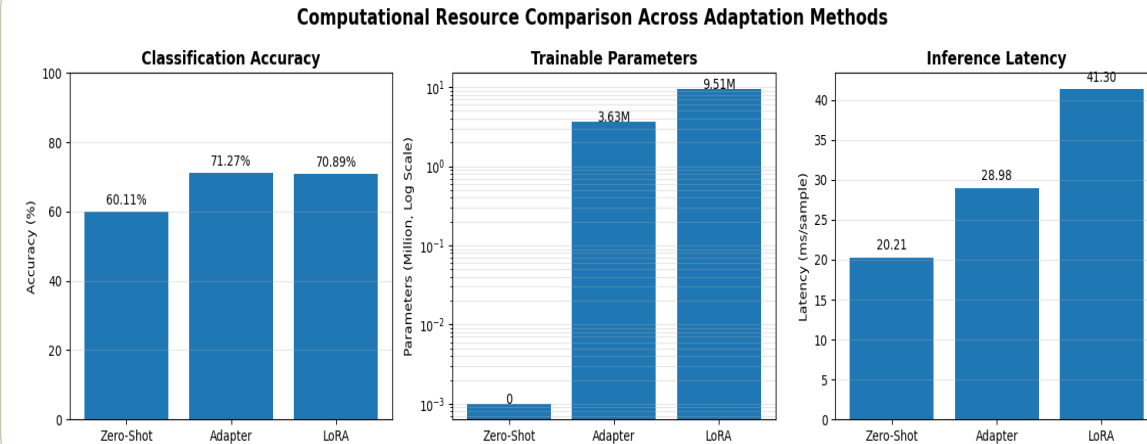


Figure :Computational Resources of Severity Classify

Adapter achieved the best accuracy, while LoRA provided a more deployment-friendly trade-off between performance and parameter efficiency. The t-SNE and examples show that clear Severe cases are easier to classify, while borderline Mild and None cases can still be confused.

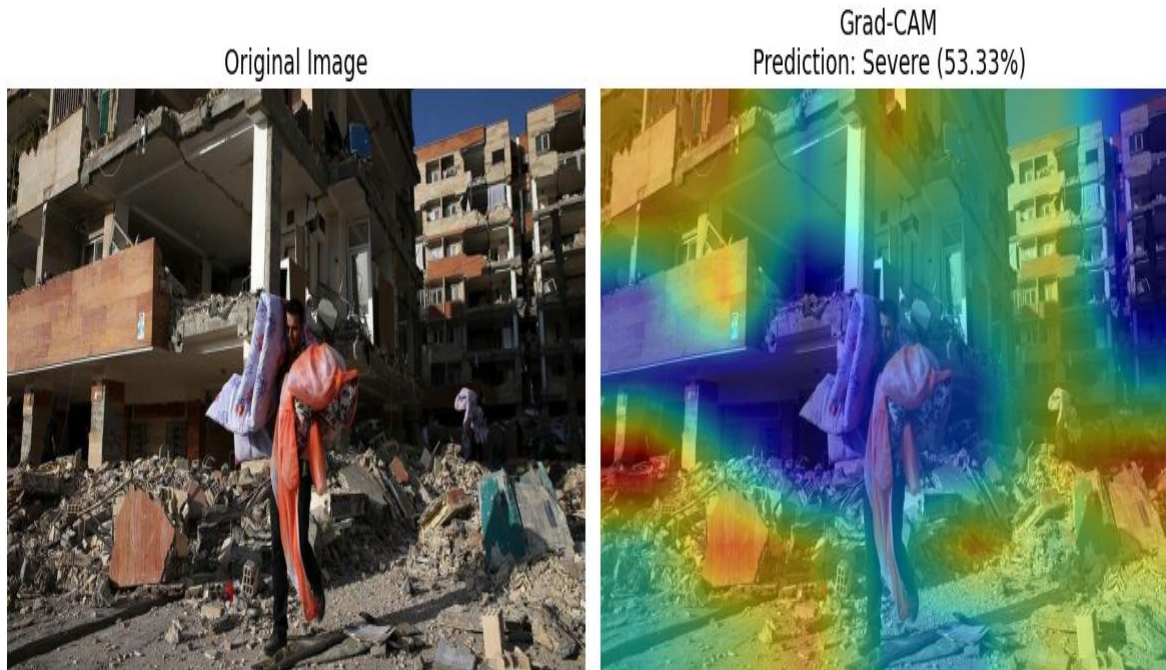


Figure: Grad-CAM for Severe Damage

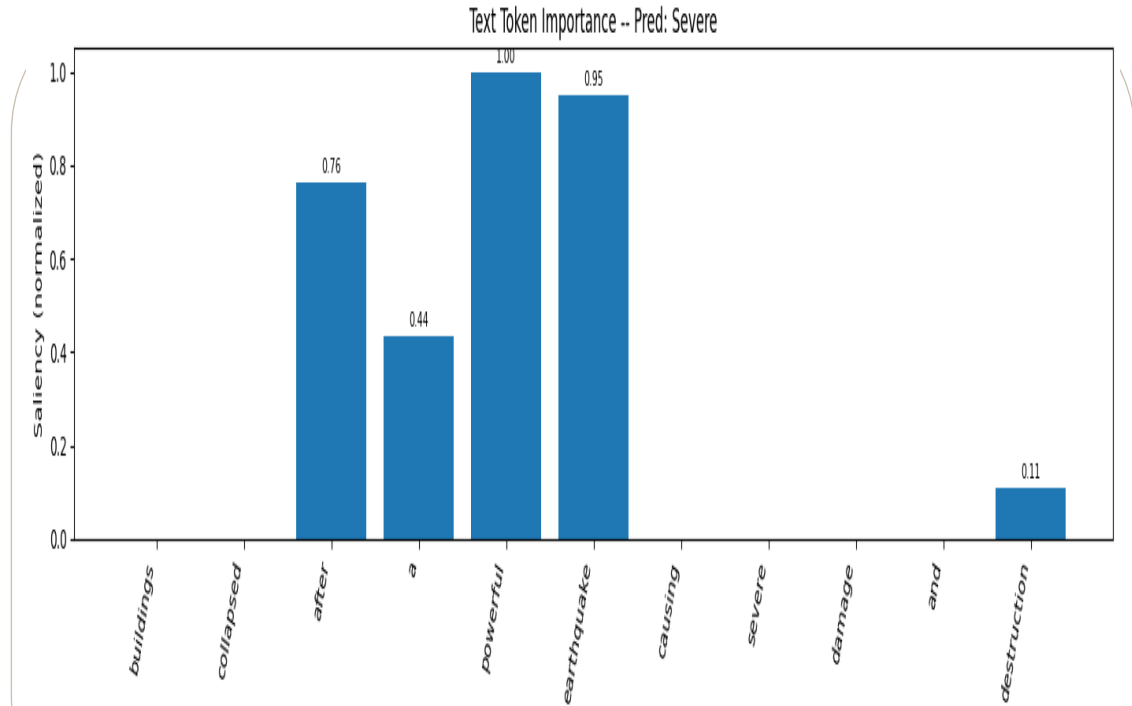


Figure :Text Token Importance of Severity Classify

Final interpretation

Grad-CAM highlights damaged structures and debris regions that contribute to severe damage prediction. Text attribution identifies disaster-related terms such as earthquake, powerful, and destruction. The explanation indicates that CRISISCLIP-X utilizes both visual and textual disaster evidence while maintaining the need for human supervision in critical decisions.

Results & Analysis(cont'd)

Streamlit Web App: Why It Was Chosen

Why Streamlit?

Fast prototype for presenting the trained model without building a complex production system.

Supports image upload and caption/tweet-text input in the same interface.

Easy to display prediction labels, confidence scores, and explanation outputs.

Suitable for thesis defense because the workflow is transparent and easy to demonstrate.

Runs locally, so the demo does not depend on external deployment or cloud hosting.

Interface Snapshot

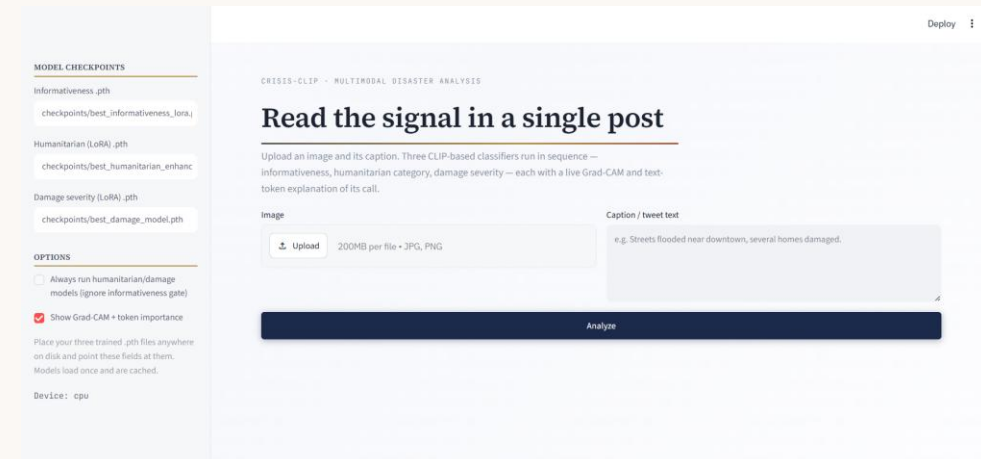
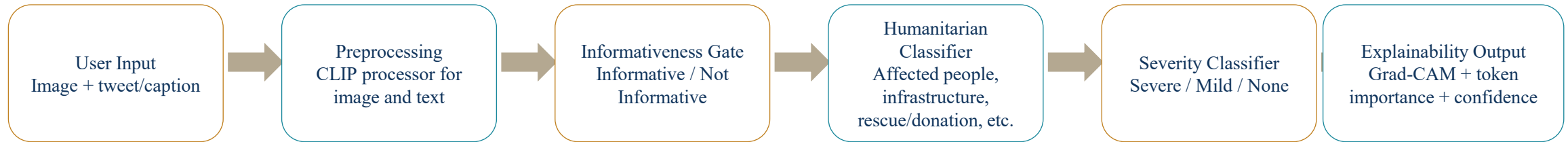


Figure : Input interface with model checkpoints, upload area, caption box, and analysis button.

Takeaway: Streamlit was selected to convert the model into an interactive, explainable, and defense-ready prototype.

Results & Analysis(cont'd)

The application follows a sequential disaster-analysis workflow: input → gate → classification → explanation.



Main Components

Sidebar: paths for three model checkpoints.
Input section: image upload and caption/tweet text area.
Analyze button: starts the sequential prediction pipeline.
Output section: class labels, confidence, and explanation visuals.

Why Sequential Design?

The informativeness model works as a first filter.
Humanitarian and severity classifiers run only when analysis is needed.
This reduces unnecessary computation and keeps the workflow understandable.
The design matches real disaster-response triage: screen first, then classify details.

Streamlit Demo Workflow

MODEL CHECKPOINTS

Informativeness .pth

checkpoints/best_informativeness_lora.j

Humanitarian (LoRA) .pth

checkpoints/best_humanitarian_enhanc

Damage severity (LoRA) .pth

checkpoints/best_damage_model.pth

OPTIONS

☐ Always run humanitarian/damage models (ignore informativeness gate)

☒ Show Grad-CAM + token importance

Place your three trained .pth files anywhere on disk and point these fields at them. Models load once and are cached.

Device: cpu

CRISIS-CLIP • MULTIMODAL DISASTER ANALYSIS

Read the signal in a single post

Upload an image and its caption. Three CLIP-based classifiers run in sequence — informativeness, humanitarian category, damage severity — each with a live Grad-CAM and text-token explanation of its call.

Image

Upload

200MB per file • JPG, PNG

Caption / tweet text

e.g. Streets flooded near downtown, several homes damaged.

Analyze

Sustainable AI Design

PEFT

Adapters and LoRA update only a small share of CLIP weights, lowering training demand.

Reuse

Pre-trained ViT-B/32 knowledge avoids costly model training from scratch.

4 GB

Memory-conscious experiments run on an RTX 3050 laptop GPU.

Trade-off

Zero-shot, Adapter and LoRA comparisons support efficiency-aware model selection.

Deploy

A lightweight Streamlit interface reduces infrastructure barriers.

Societal & Practical Impact

FAST

Rapidly screens disaster images and text at scale.

PRIORITY

Highlights severity, affected conditions and humanitarian needs.

ACTION

Turns unstructured posts into actionable information for response teams.

TRUST

XAI explanations make predictions easier to inspect and interpret.

USABLE

A simple interface supports use by non-technical personnel.

Real-World Application Workflow

Social media
disaster data



CrisisMMD +
MEDIC
analysis



CRISISCLIP-X
classification



Actionable disaster
information



Emergency planning &
resource management

Core contribution: efficient multimodal AI that supports faster, more transparent and accessible disaster-response decisions.

Evaluation Aspect	Existing Approaches	CRISISCLIP-X Result	Advantage
Humanitarian Classification	CNN + Word2Vec + VGG16 F1 = 78.3% FastText + VGG16 F1 $\approx$ 87.5%	Adapter CLIP Accuracy = 90.58% Macro-F1 = 0.8324	Improved classification using pretrained vision–language representation
Informativeness Classification	CNN + VGG16 F1 = 84.2% GPT-2 + ViT-B/16 Accuracy = 84.66%	LoRA CLIP Accuracy = 87.17% Weighted F1 = 87.15%	Higher performance with efficient parameter updates
Parameter Efficiency	Conventional full-model training	LoRA updates only 1.03% of trainable parameters	Reduced computational cost and better deployment feasibility
Explainability	Most existing methods provide only predictions	Grad-CAM + text token attribution + XAI analysis	Provides interpretable visual and textual evidence
Task Coverage	Mostly a single disaster-classification task	Three tasks: Informativeness • Humanitarian Category • Severity	Unified multimodal disaster-analysis framework

LIMITATIONS

1

Limited and Imbalanced Data

CrisisMMD and MEDIC cover limited disaster types, locations, languages, and time periods. Minority classes remain harder to learn.

2

Noisy Multimodal Content

Social-media posts may contain blurred images, missing files, duplicate content, short captions, hashtags, or misinformation. Image and text may also conflict.

FUTURE AWORK

3

Broader Validation

Expand datasets across disaster types, regions, languages, seasons, and minority classes; validate with real field data.

4

Robust Multimodal AI

Improve handling of missing, noisy, or conflicting modalities and test stronger fusion and XAI-faithfulness methods.

5

Efficient Edge Deployment

Use quantisation, pruning, distillation, and hardware-aware optimisation for embedded GPU or FPGA deployment.

6

Real-Time Response System

Integrate sensors, GPS, live dashboards, communication networks, and human supervision for responsible deployment.

Future direction: develop CRISISCLIP-X into an efficient, explainable, human-supervised system for reliable real-time disaster response.



Figure : Publication of this reasearch

References (cont'd)

- [1] M. Imran, C. Castillo, F. Diaz, and S. Vieweg, "Processing social media messages in mass emergency: Survey summary," in *WWW Companion*, 2018. [Online]. Available: <https://doi.org/10.1145/3184558.3186242>
- [2] C. Castillo, *Big Crisis Data: Social Media in Disasters and Time-Critical Situations*. Cambridge Univ. Press, 2016. [Online]. Available: <https://doi.org/10.1017/9781316476840>
- [3] L. Palen and S. B. Liu, "Citizen communications in crisis: Anticipating ICT-supported public participation," in *CHI*, 2007. [Online]. Available: <https://doi.org/10.1145/1240624.1240736>
- [4] S. Vieweg, A. L. Hughes, K. Starbird, and L. Palen, "Microblogging during two natural hazards events," in *CHI*, 2010. [Online]. Available: <https://doi.org/10.1145/1753326.1753486>
- [5] M. Imran, C. Castillo, J. Lucas, P. Meier, and S. Vieweg, "AIDR: Artificial intelligence for disaster response," in *WWW Companion*, 2014. [Online]. Available: <https://doi.org/10.1145/2567948.2577034>
- [6] D. T. Nguyen, F. Ofli, M. Imran, and P. Mitra, "Damage assessment from social media imagery data during disasters," in *ASONAM*, 2017. [Online]. Available: <https://doi.org/10.1145/3110025.3110109>
- [7] F. Alam, F. Ofli, and M. Imran, "CrisisMMD: Multimodal Twitter datasets from natural disasters," in *ICWSM*, 2018. [Online]. Available: <https://ojs.aaai.org/index.php/ICWSM/article/view/14983>
- [8] F. Alam, F. Ofli, and M. Imran, "CrisisMMD dataset resource," 2018. [Online]. Available: <https://crisisnlp.qcri.org/crisismmd>
- [9] S. Gite *et al.*, "Analysis of multimodal social media data utilizing ViT Base 16 and GPT-2 for disaster response," *Arabian J. Sci. Eng.*, 2025. [Online]. Available: <https://doi.org/10.1007/s13369-025-10314-7>
- [10] H. Mouzannar, Y. Rizk, and M. Awad, "Damage identification in social media posts using multimodal deep learning," in *ISCRAM*, 2018. [Online]. Available: https://idl.iscram.org/files/husseinmouzannar/2018/2129_HusseinMouzannar_etal2018.pdf

References (cont'd)

- [11] M. Agarwal, M. Leekha, R. Sawhney, and R. R. Shah, "Crisis-DIAS: Towards multimodal damage analysis - Deployment, challenges and assessment," in *AAAI*, 2020, pp. 346–353. [Online]. Available: <https://doi.org/10.1609/aaai.v34i01.5369>
- [12] F. Alam, F. Ofli, M. Imran, T. Alam, and U. Qazi, "Deep learning benchmarks and datasets for social media image classification for disaster response," in *ASONAM*, 2020, pp. 151–158. [Online]. Available: <https://doi.org/10.1109/ASONAM49781.2020.9381294>
- [13] F. Ofli, F. Alam, and M. Imran, "Analysis of social media data using multimodal deep learning for disaster response," in *ISCRAM*, 2020. [Online]. Available: <https://arxiv.org/abs/2004.11838>
- [14] Z. Zou, H. Gan, Q. Huang, T. Cai, and K. Cao, "Disaster image classification by fusing multimodal social media data," *ISPRS Int. J. Geo-Inf.*, vol. 10, no. 10, p. 636, 2021. [Online]. Available: <https://doi.org/10.3390/ijgi10100636>
- [15] F. Alam, T. Alam, M. A. Hasan, A. Hasnat, M. Imran, and F. Ofli, "MEDIC: A multi-task learning dataset for disaster image classification," *Neural Comput. Appl.*, vol. 35, no. 3, pp. 2609–2632, 2023. [Online]. Available: <https://doi.org/10.1007/s00521-022-07717-0>

Thank You
